

Dhruv Sagar

dhruvsagar.pro@gmail.com

github.com/dhruvel
linkedin.com/in/dhruvel
dhruvel.netlify.app

SKILLS

Programming Languages: C, C++, Go, Java, JavaScript/TypeScript, LaTeX, Markdown, Python, Rust, Solidity, SQL

Frameworks/Libraries: Apple Metal, Keras, Matplotlib, Node.js, Numpy, OpenCV, OpenGL, Pandas, ReactJS, React Native, Scipy, Selenium, Sklearn, Tensorflow

Technologies: AWS, Android Studio, Claude, Docker, Git, Github Copilot, GCP, HomeAssistant, Jenkins, Kubernetes, MySQL, PostgreSQL

AI: Agent Skills, Agents, Claude, Logging, MCP, RAG, Roots, Sampling, Skills, STDIO/StreamableHTTP, Sub-agents, Tools

CERTIFICATIONS

Anthropic Courses | *Skilljar*

July 2026

Agent Skills

Credential ID kvu3mcug2jm6

MCP

Credential ID tns865e26v4u

MCP: Advanced

Credential ID 7o4uc2d6s7ka

Subagents

Credential ID f9ixtbfbfqvm

EXPERIENCE

Robotics Software Engineer | *Peribott Dynamics LLP*

May 2025 – April 2026

Hyderabad, India

- Engineered a schema driven ROS 2 SLAM pipeline integrating ASCAM HP60C RGBD camera and LDROBOT D500 LiDAR with RTABMap, achieving 20Hz frame processing on 640×480 resolution camera feed.
- Developed sensor fusion pipeline combining RGBD imaging and LiDAR data with 5 sensor real time monitoring (2s reporting interval), configurable ICP registration (0.3 correspondence ratio, 0.02m voxel size), and comprehensive diagnostic tools enabling sub 5sec detection of sensor failures.
- Created reusable launch configurations with optimized parameters across 30 frames short term memory, 5m local radius, and dual odometry tracking (0.001 linear/angular variance), plus Python sensor validators and performance tuning templates that standardized SLAM deployment across multiple robot platforms.

Junior Accountant / Operations Assistant | *PND Complex*

November 2023 – April 2025

Hyderabad, India

- Managed financial operations including accounts payable/receivable, payroll processing, and budget tracking for a small size employee organization, ensuring compliance with local tax regulations and financial reporting standards.
- Implemented an automated invoicing system using Notion, n8n, and WhatsApp, reducing manual data entry time by 40% and improving accuracy in financial records.
- Assisted in the development of operational workflows, streamlining communication between departments and enhancing overall efficiency in everyday business activities.

Beta Videogame Tester | *Testify*

February 2020 – April 2022

Remote, Canada

- Conducted comprehensive beta testing for multiple video game titles, identifying and documenting bugs, glitches, and gameplay issues to provide actionable feedback to development teams.
- Collaborated with cross functional teams to ensure timely resolution of reported issues, contributing to the overall quality and user experience of the final product.
- Utilized various testing tools and platforms to simulate different gaming environments and scenarios, ensuring thorough coverage of potential user experiences.

EDUCATION

Wilfrid Laurier University

Waterloo, Canada

Honours Bachelor of Science; with Distinction

January 2020 – December 2023

- Major:** Computer Science
- Minor:** Big Data Systems
- Concentration:** Artificial Intelligence & Machine Learning

TECHNICAL PROJECTS

[Blockchain Based DNS Protection System](#) | *Python, Ethereum, Web3.py, Solidity*

- Deployed Ethereum smart contract serving DNS queries with configurable TTLs (300s standard, 86,400s NS/SOA) over UDP/TCP port 53 with 5s query timeout.
- Achieved 167ms blockchain lookup vs 46ms traditional DNS, with instant DNS updates (vs 7 day propagation) via Sepolia testnet smart contract and role based access control.
- Implemented distributed domain storage with 14 DNS record types, 4 sanitization algorithms, and domain expiration tracking to eliminate cache poisoning attacks.

[Data Dissolve](#) | *Java, Android SDK*

- Implemented 4 cryptographic file deletion algorithms (Gutmann 35pass, DoD 7pass, Schneier 7pass, Custom Npass) to permanently overwrite sensitive data on Android devices.
- Architected a schema driven Android application with PIN authentication, multi language support (3 languages), and intuitive UI for file/folder secure deletion and free space wiping.
- Engineered efficient device memory management through secure data purging while reducing fragmentation and improving overall device performance.